

Exercises on

Quantum Field Theory

(Based on the lectures by *Dr. M Arshad Momen*)

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1 Exercise for Class-2 [Lecture-1]

Derive the Euler–Lagrange equations corresponding to this action:

$$S[x] = m \int \sqrt{\eta_{ab} dx^a dx^b} = m \int d\tau = m \int \sqrt{\eta_{ab} \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda}} d\lambda$$

2 Exercise for Class-4 [Lecture-1]

2.1 Assignment-1

How does $\hat{J}^{rs}$ transform under Poincaré transformation? (It's a semi direct product)

2.2 Assignment-2

Recover 3D rotation algebra from angular momentum.

3 Exercise for Class-5 [Lecture-2]

3.1 Exercise-1

Show that we can recover the Lorentz force from the action:

$$S = \frac{1}{2} \int m \dot{\xi}^2 d\tau + q \int A_b(\xi) \dot{\xi}^b d\tau$$

[This latter interaction term is geometrical as well]

3.2 Exercise-2

Check the commutations:

$$\begin{aligned} [J, p] &\sim p \\ [p, p] &\sim 0 \\ [J, J] &\sim J \\ [J, p^2] &\sim 0 \\ [J, p_a p^a] &= [J, p_a] p^a + p_a [J, p^a] \sim 0 \end{aligned}$$

4 Exercise for Class-6 [Lecture-3]

4.1 Exercise-1

Show that the lie derivatives of vector field satisfies:

$$\mathfrak{L}_{\hat{V}} \mathfrak{L}_{\hat{W}} - \mathfrak{L}_{\hat{W}} \mathfrak{L}_{\hat{V}} = \mathfrak{L}_{[\hat{V}, \hat{W}]}$$

4.2 Exercise-2

Derive the lie derivative for a tensor $\mathfrak{L}_{\hat{V}} \tilde{T}^a{}_b$.

5 Exercise for Class-7 [Lecture-4]

- Calculate $\mathfrak{L} g_{ab}$

Hint : Let choose,

$$g_{ab} = P_a Q_b$$

$$\begin{aligned}\mathfrak{L}_v(P_a Q_b) &= (\mathfrak{L}_v P_a) Q_b + P_a (\mathfrak{L}_v Q_b) \\ \mathfrak{L}_v g_{ab} &= V^c \partial_c g_{ab} + g_{ac} \partial_b V^c + g_{cb} \partial_a V^c\end{aligned}$$

- Verify if the partial derivative is switched with the covariant derivative ($\partial \rightarrow \nabla$) the Lie derivative won't change.

6 Exercise for Class-8 [Lecture-5]

Derive the equation of motion for this action and compare with the KG equation to determine the constants -

$$S[\phi] = \alpha \phi_i \phi_i + \beta \partial \phi_i \partial \phi_i + \gamma_i \phi_i + K_i \partial \phi_i$$

Note: The last term is important for π -meson.

7 Exercise for Class-11 [Lecture-8]

Find the solution of the $G(t)$.

8 Exercise for Class-13 [Lecture-10]

Do this calculation for 6-point function and calculate $\langle \phi_a \phi_b \phi_c \phi_d \phi_e \phi_f \rangle$.

9 Exercise for Class-14 [Lecture-11]

9.1 Exercise-1

Consider a field theory for two fields ϕ, χ and whose action is given by

$$S = \frac{1}{2} \phi^T A \phi + \frac{1}{2} \chi^T A \chi + \chi^T B \phi - \phi^T B \chi \quad , \quad [B = -B]$$

Find the propagators for this field.

9.2 Exercise-2

For $V = \frac{g\phi^3}{3!}$, Calculate the 2nd order contribution to $Z[J]$ due to this interaction.

10 Exercise for Class-15 [Lecture-12]

Decompose the full 4-point function in terms of the connected Green's function of 4th order or less.

11 Exercise for Class-16 [Lecture-13]

Decompose the 4-point connected Green function in terms of proper vertices and the 2-point function.

12 Exercise for Class-19 [Lecture-16]

For n such vertex insertions: each minus comes from the propagator in a Euclideanized way,

$$(-1)^n \int \frac{d^D k}{(2\pi)^D} \frac{[V''(\varphi)]^n}{(k^2 + m^2)^n}.$$

Using dimensional regularization,

$$\int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 + m^2)^n} = \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(n - \frac{D}{2})}{\Gamma(n)} \frac{1}{(m^2)^{n-D/2}}.$$

Thus,

$$= (-1)^n \frac{[V''(\varphi)]^n}{(4\pi)^{D/2}} \frac{1}{\Gamma(n)} \frac{\Gamma(n - \frac{D}{2})}{(m^2)^{n-D/2}}.$$

So,

$$\Gamma[\varphi] = \frac{\hbar}{2i} \sum_{n=1}^{\infty} \left[-\frac{V''(\varphi)}{m^2} \right]^n \frac{\Gamma(n - \frac{D}{2})}{\Gamma(n)} (m^2)^{D/2}.$$

HW: Resume this series (the first terms) to obtain a closed expression.

13 Exercise for Class-20 [Lecture-17]

13.1 Assignment-1

Redo today's calculation for $D = 3, 6$.

13.2 Assignment-2

Fix the today's μ factor.

13.3 Assignment-3

Express V_T in terms of m^{-2} and ϕ .

13.4 Assignment-4

$$\frac{d\Gamma_1}{dm^2} = \frac{\hbar}{16\pi^2} \int_0^{\Lambda^2} \frac{z}{z + m^2} dz = \frac{\hbar}{16\pi^2} \int_0^{\Lambda^2} \left[1 - \frac{m^2}{z + m^2} \right] dz.$$

Perform the integral explicitly.

13.5 Assignment-5

Check that

$$[Q^a, Q^b] = if^{abc}Q^c$$

13.6 Assignment-6

$$\int d\eta_1 \dots d\eta_N e^{\eta^T A \eta} = \det(A)$$

By induction method:

$$\eta^T = -\eta, \quad A^T = -A$$